GGG-N-350a

JUNE 8, 1959

SUPERSEDING
Int. Fed. Spec. GGG-N-00350 (Navy-Ships)
April 29, 1957 and
Fed. Spec. GGG-P-471a (in part)
June 26, 1953

FEDERAL SPECIFICATION

NIPPERS AND PINCERS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers end cutting nippers and pincers (plier-like hand tools having two opposing sharp-edged jaws, each at right angles to the axis of the handles, actuated on a pivot to develop a cutting action), used in assembly or repair work.

1.1.1 *Federal specification coverage.*—This specification does not include all types of nippers and pincers which are commercially available, but includes only those types generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types, classes, and styles.*—Nippers and pincers shall be of the following types, classes, and styles as specified (see 6.1):

Type I.—Nippers, end cutting.
- Class 1.—Detachable cutters.
  - Style A.—Rounded.
  - Style B.—Flat.
  - Style C.—Flat, adjustable jaws.
- Class 2.—Integral cutter.
  - Style A.—Normal duty.
  - Style B.—Light duty (jewelers').

Type II.—Pincers.
- Class 1.—Farriers'.
- Class 2.—Carpenters'.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

UU-P-121—Paper, Bond and Writing, White and Colored.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

FSC 5110, 5120

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Military Specifications:

MIL–P–15424—Packaging of Hand Tools for Domestic and Overseas Shipment and Storage.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

MIL–STD–130 — Identification Marking of U. S. Military Property.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Illustrations.—The illustrations herein are descriptive and not restrictive and are included for the convenience of requisitioning and purchasing officers and manufacturers, and are not intended to preclude the purchase of nippers and pincers otherwise conforming to this specification.

3.2 Material.—Type I, class 1 nippers shall be of malleable iron or steel. Type I, class 2 nippers and type II pincers shall be forged from steel of a quality which, when properly heat-treated, shall comply with the applicable requirements.

3.3 Design.—Nippers and pincers shall be properly proportioned in all parts so as to be strong, durable and easy to operate.

3.4 Handles. — Handles of nippers and pincers shall be so shaped as to afford a comfortable grip and shall be free from rough edges and sharp corners. Inner handle surfaces adjacent to the joint shall be well-rounded or beveled and may have metal stops so as to prevent accidental pinching of the hands. Ends of handles of nippers and pincers shall not touch when the jaws are in the closed position. Outer hand-gripping surfaces of handles shall be smooth, knurled, or impressed.

3.5 Joints.—Nippers and pincers shall have either lap or box joints. There shall be no excessive sidewise movement, play or other indication of looseness of the two halves of the tool when the nippers or pincers are open or closed.

3.6 Jaws.

3.6.1 *Jaw openings.*—The jaws of nippers and pincers shall open to the respective minimum distance specified for the individual types, and sizes, through which distance the tool shall operate in a smooth and uniform manner. It shall be possible to open and close the handles through the minimum jaw opening specified in the applicable table by the action of one hand only. Beyond the minimum jaw-opening distance, the jaws may open at increased loads until the positive stop of the tool is engaged.

3.6.2 *Jaw hardness.* — Unless otherwise specified herein, the jaws of nippers and pincers, adjacent to the cutting edge, shall show a hardness value of not less than 55 and not more than 65 on the Rockwell "C" scale when tested as specified in 4.4.1.

3.7 Finishes.

3.7.1 *Surfaces.* — Surfaces shall be free from roughness, sharp edges, and corners, and excess oxide scale.

3.7.2 *Coatings.*—Unless otherwise specified (see 6.1), nippers and pincers shall have one coating or a combination of the following coatings:

(a) Natural, oil treated.

(b) Chemically produced oxide coating, oil treated.

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(c) Chemically produced phosphate coating, oil treated.
(d) Polished all over, oil treated.
(e) Electrodeposited metallic coating.
(f) Polished heads, oil treated.
(g) Painted.

3.8 Identification of product.—Nippers and pincers shall be marked in accordance with Military Standard MIL–STD–130, as practicable.

3.9 Type I, nippers, end cutting.

3.9.1 *Class 1, detachable cutters.*—Class 1 nippers shall be suitable for heavy-duty cutting service. The upper and lower cutter members shall be well-proportioned and made of properly hardened tool steel, readily removable from the jaws for grinding or renewal. Cutting edges shall be parallel to each other and properly aligned. There shall be a substantial steel stop screw fitted at the back of the joint, readily adjustable to prevent the cutting edges from closing on each other. Class 1 nippers shall be capable of withstanding the handle load and steel-wire-cutting tests of 4.4.3.1.1 and 4.4.3.1.2 respectively.

3.9.1.1 *Style A, rounded.*—Style A nippers shall conform to table I for the size specified (see 6.1), and shall be similar to figure 1.

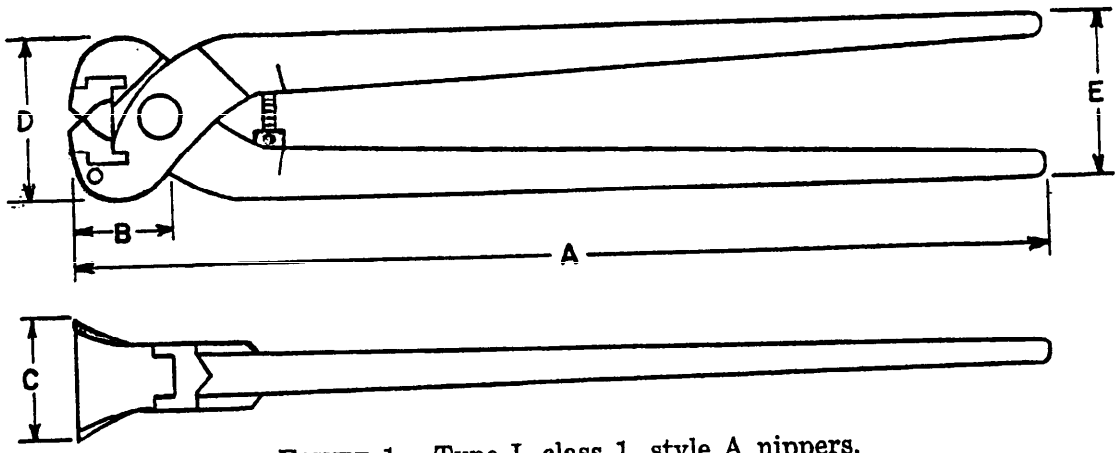


FIGURE 1.—Type I, class 1, style A nippers, end cutting, detachable cutters, rounded.

TABLE I.—*Type I, class 1, style A nippers, end cutting, detachable cutters, rounded*

Nominal size	Overall length A ±½	Center of rivet to end of jaw B ±⅛	Thickness of jaw C ±⅛	Width of jaw D ±5/16	Span of handles in closed position E ±5/16	Jaw opening, minimum	Weight, maximum
Inches	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Ounces*
10	11	1⅛	15/16	2½	2¼	¼	32
12	13	1⅛	1 1/16	2¾	2⅜	¼	42
14	15	1⅛	1½	2⅞	2½	⅜	64

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3.9.1.2 *Style B, flat (see 6.2).*—Style B nippers shall conform to table II for the size specified (see 6.1), and shall be similar to figure 2.

3.9.1.3 *Style C, flat, adjustable jaws (see 6.2).*—Style C nippers shall be made of drop forged steel and shall be of compound leverage construction with box joints. The finish shall be in accordance with 3.7.2(d). The detachable jaws shall be made of a high carbon alloy tool steel and shall show a hardness value of not less than 60 and not more than 63 on the Rockwell "C" scale when tested as specified in 4.4.1. Each jaw shall have a minimum adjustment of ¼ inch. Each jaw shall be secured to the frame by two screws, extending through the jaws and engaging a sliding threaded member in a slot in the frame. The mating surfaces of the jaws and frame shall have machine cut teeth parallel to the cutting edges of the jaws. The teeth shall be spaced to provide adjustment in increments of 0.050 inch or less. The handles shall employ a compression spring which holds the jaws apart when not being used. When the stop screw and jaws are set so that the widest handle span is 2 1/16 inches in the closed position, each jaw shall be capable of being adjusted a minimum of ¼ inch to compensate for resharpening. A flat flexible spring shall be incorporated approximately 5/16 inch below the cutting edges of the jaws to provide a yielding seat for the end of cut wire to minimize the possibility of chipping the cutting edges when heavy wire is cut. Close tolerances on the fitting of pins, rivets, and box joints shall be maintained to eliminate undesirable looseness in the moving sections. Style C nippers shall conform to table III and shall be similar to figure 3.

3.9.2 *Class 2, integral cutter.*—The cutting edges of class 2 nippers shall be integral with

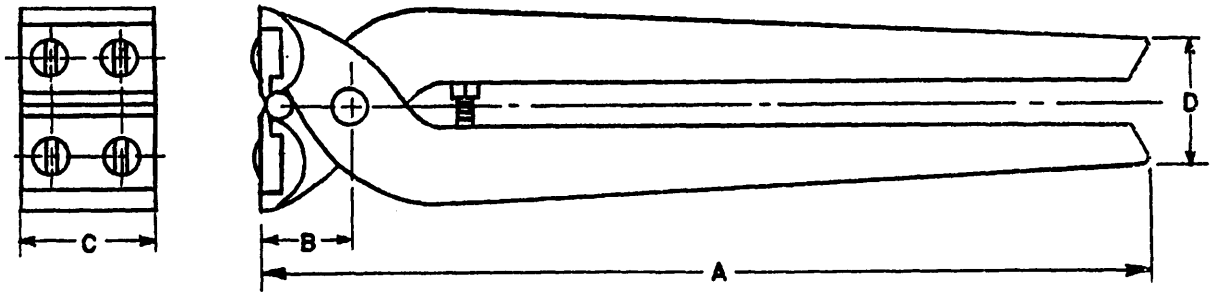


FIGURE 2.—Type I, class 1, style B nippers, end cutting, detachable cutters, flat.

TABLE II.—*Type I, class 1, style B nippers, end cutting, detachable cutters, flat*

Nominal size	Overall length A		Center of rivet to end of jaw B ±1/8	Thickness of jaw C ±1/8	Span of handles in closed position D ±3/16	Jaw opening, minimum	Weight, maximum
	Minimum	Maximum					
Inches	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inch*	*Ounces*
6	5½	6½	15/16	1	1¾	¼	10
10	9½	10½	15/16	1½	1 13/16	5/16	30
12	11½	12½	15/16	1¾	1 13/16	5/16	38
14	14	15	1 3/16	1⅞	1 13/16	⅜	62

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the jaw and at right angles to the plane of the handles. The cutting edges shall contact throughout their entire length when the nippers are closed.

3.9.2.1 *Style A, normal duty.*—Style A nippers shall be suitable for cutting wire close to flat or concave surfaces and shall be capable of withstanding the tests of 4.4.3.2.1.1, 4.4.3.2.1.2, and 4.4.3.2.1.3, respectively. Style A nippers shall conform to table IV for the size specified (see 6.1) and shall be similar to figure 4.

3.9.2.2 *Style B, light duty (jewelers').*—Style B nippers shall be suitable for light duty work such as cutting fine wire and for use in jewelry work. Nippers shall be of nickel alloy steel with polished jaws. Style B nippers shall be capable of withstanding the handle-load test of 4.4.3.2.2. Style B nippers shall conform to table V and shall be similar to figure 5.

3.10 **Type II, pincers.**—The cutting edges of type II pincers shall be at right angles to the plane of the handles and shall contact throughout their entire length when the tool is closed.

3.10.1 *Class 1, farriers'.*—Class 1 pincers shall be suitable for farriers' use. Handles shall be straight throughout. Class 1 pincers shall be capable of withstanding the tests specified in 4.4.4.1. Class 1 pincers shall conform to table VI for the size specified (see 6.1), and shall be similar to figure 6.

3.10.2 *Class 2, carpenters'.*—The pincers shall be suitable for use in carpentry work and shall be of high grade, oil-tempered tool steel. One handle shall have a screwdriver end for use on screws and for prying. The other handle shall have an integrally forged "Vee"-shaped opening at the end for extracting tacks or small nails. The pincers shall be capable of withstanding the tests of 4.4.4.2.1 and 4.4.4.2.2, respectively. Class 2 pincers shall conform to table VII for the size specified (see 6.1), and shall be similar to figure 7.

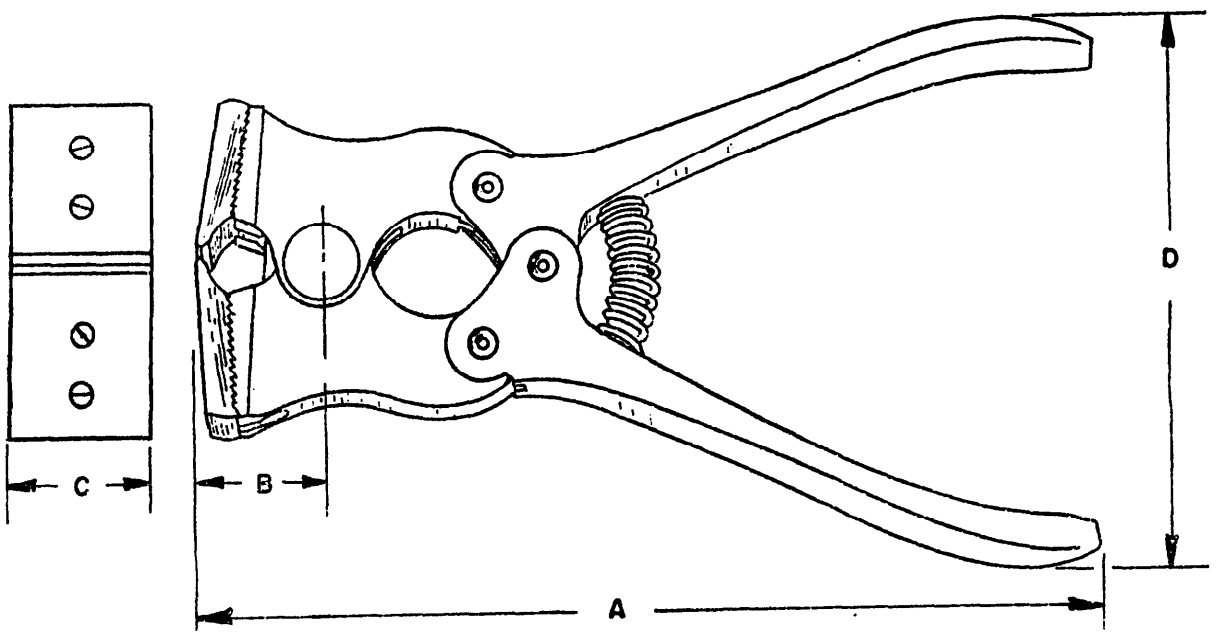


FIGURE 3.—Type I, class 1, style C nippers, end cutting, adjustable jaws, flat.

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TABLE III.—*Type I, class 1, style C nippers, end cutting, adjustable jaws, flat*

Nominal size	Overall length A		Center of rivet to end of jaw B ±1/8	Thickness of jaw C ±1/8	Span of handles in closed position D ±5/16	Jaw opening, minimum	Weight, maximum
	Minimum	Maximum					
Inches	*Inches*	*Inches*	*Inch*	*Inch*	*Inches*	*Inch*	*Ounces*
7	6 7/8	7 1/8	13/16	13/16	2 1/16	3/32	20

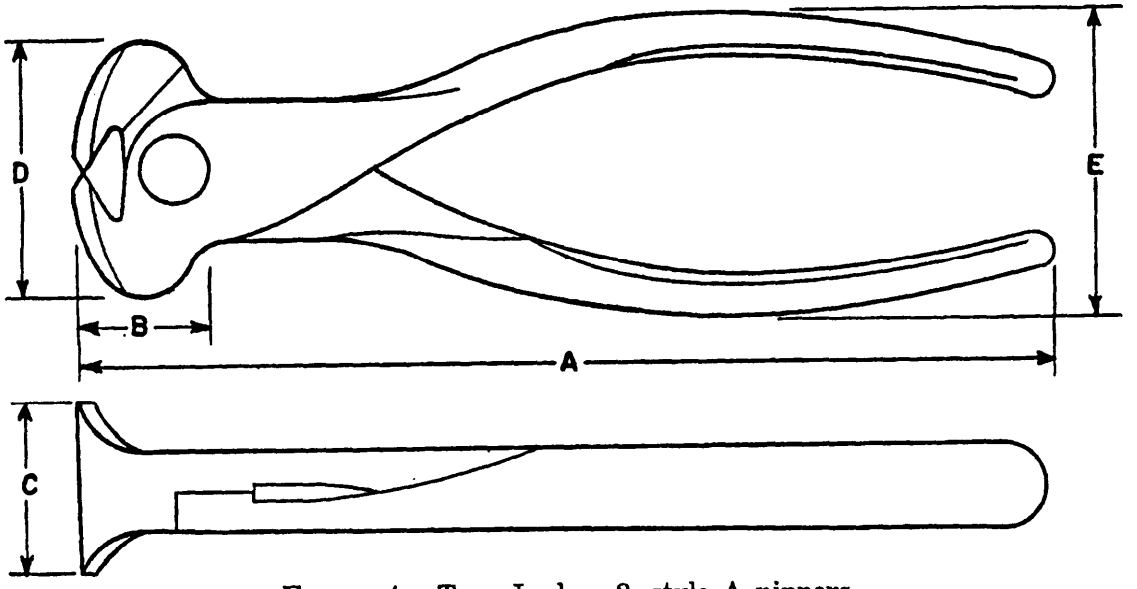


FIGURE 4.—Type I, class 2, style A nippers, end cutting, integral cutter, normal duty

TABLE IV.—*Type I, class 2, style A nippers, end cutting, integral cutter, normal duty*

Nominal size	Overall length A ±1/2	Center of rivet to end of jaws B ±5/32	Thickness of jaws C ±1/4	Width of jaws D ±1/8	Span of handles in closed position E ±5/16	Jaw opening, minimum	Weight, maximum
Inches	*Inches*	*Inch*	*Inches*	*Inches*	*Inches*	*Inch*	*Ounces*
4 1/2	4 1/2	1/2	3/4	5/8	1 5/8	1/8	8
6	6 1/4	11/16	1 1/16	1 1/2	1 3/4	3/16	10
7	7 1/4	3/4	1 3/16	1 11/16	1 7/8	1/4	14

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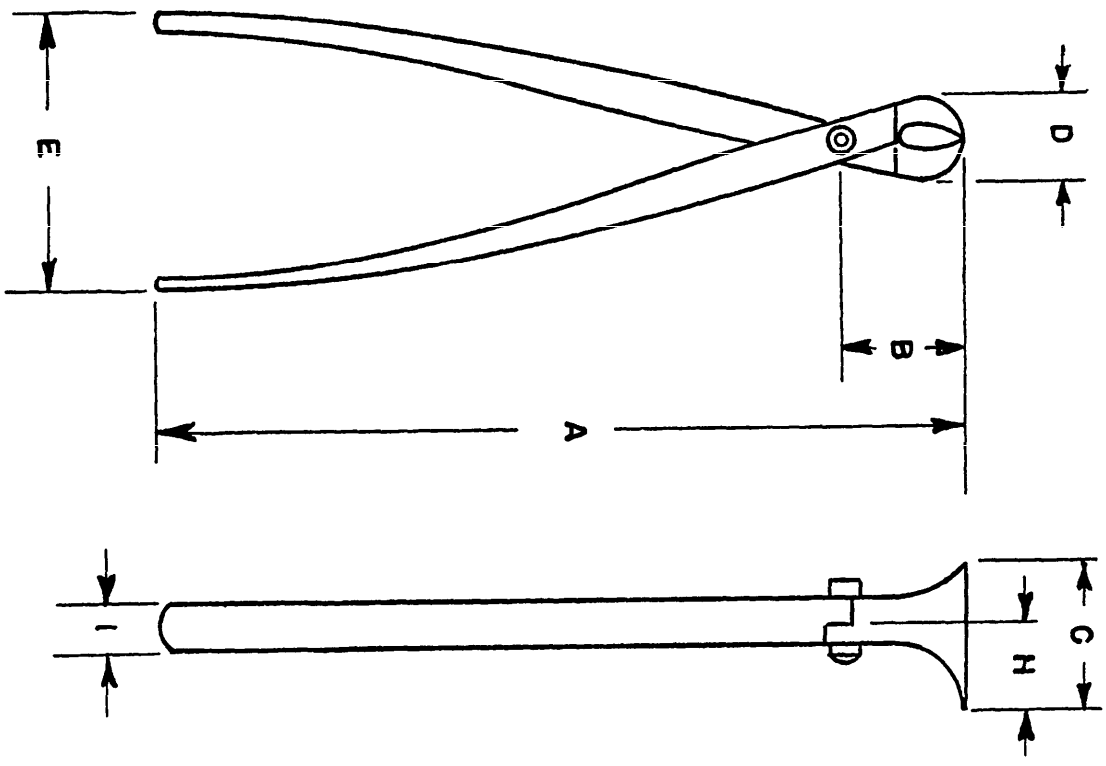


FIGURE 5.—Type I, class 2, style B nippers, end cutting, integral cutter, light duty (jewelers').

TABLE V.—*Type I, class 2, style B nippers, end cutting, integral cutter, light duty (jewelers')*

Nominal size	Overall length A ±1/8	Center of rivet to end of jaws B ±1/64	Thickness of jaws C ±1/64	Width of jaws D ±1/64	Span of handles in closed position E ±1/8	Long side of jaw H	Width of handle I ±1/16	Jaw opening, minimum	Weight, maximum
Inches	*Inches*	*Inch*	*Inch*	*Inch*	*Inches*	*Inch*	*Inch*	*Inch*	*Ounces*
4½	4½	5/8	11/16	3/8	1 7/16	3/8	1/4	1/8	2

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TABLE VI.—*Type II, class 1, pincers, farriers'*

Nominal size	Overall length A ±½	Center of rivet to end of jaw B ±⅛	Thickness of jaw C ±¼	Width of jaws D ±¼	Span of handle in closed position E ±1/16	Width of handle F ±1/16	Inside span with jaws in closed position G ±¼	Thickness of handle H ±⅛	Rivet body diameter J ±1/16	Depth of jaw K ±¼	Jaw opening, (min.)	Weight, (max.)
Inches	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inch*	*Inches*	*Inch*	*Inch*	*Inch*	*Inches*	*Ounces*
8	8	1 3/16	¾	1 5/8	2	⅜	1 3/16	5/16	5/16	9/16	15/16	14
10	10	1¼	⅞	1⅞	2 3/8	7/16	1 5/16	5/16	⅜	⅝	15/16	22
12	12	1½	1 1/16	2 1/16	2 3/8	15/32	1⅜	5/16	7/16	⅞	1⅛	34
14	14	1¾	1⅛	2¾	2½	½	1⅜	5/16	7/16	1	1¼	40

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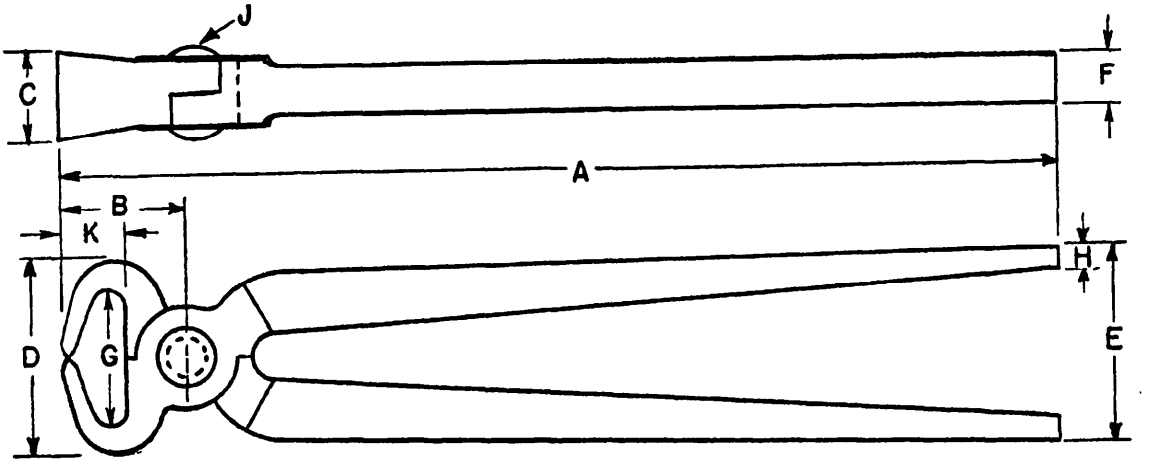


FIGURE 6.—Type II, class 1, pincers, farriers'.

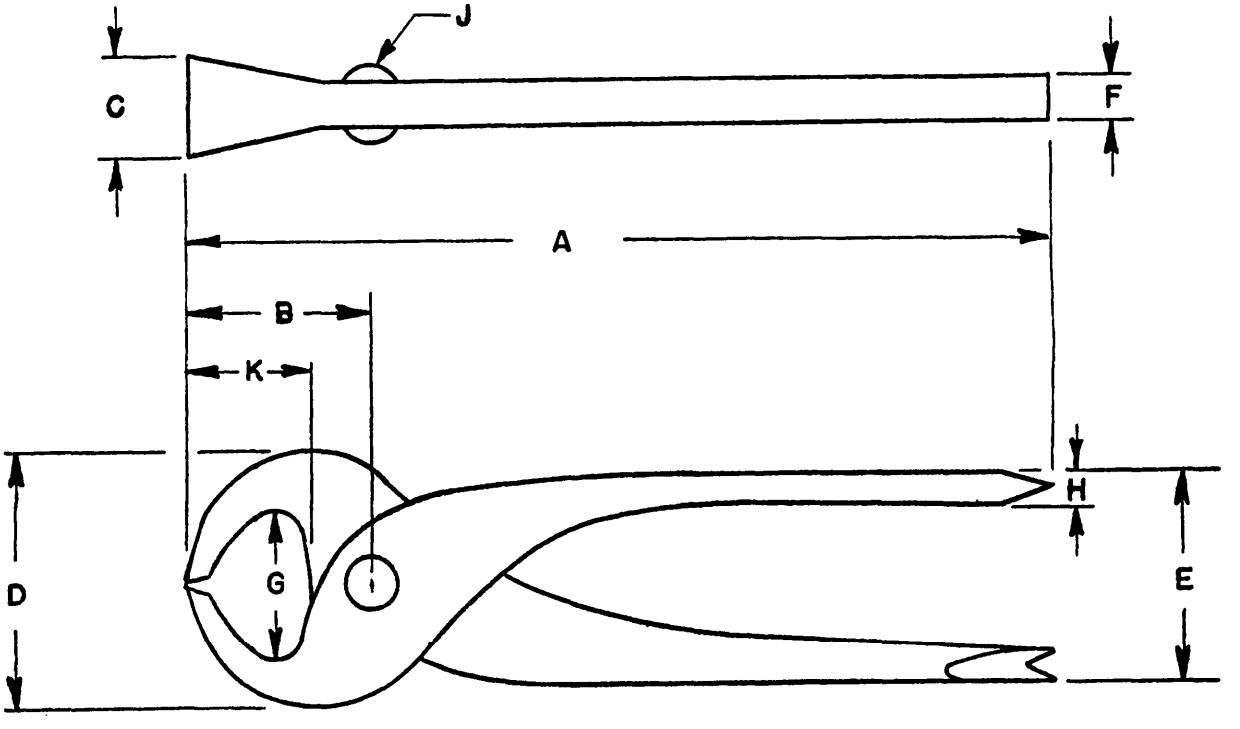


FIGURE 7.—Type II, class 2, pincer, carpenters'.

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TABLE VII.—*Type II, class 2, tinners, carpenters'*

Nominal size	Overall length A ±1/2	Center of rivet to end of jaw B ±1/8	Thickness of jaw C ±1/4	Width of jaw D ±1/4	Span of handle in closed position E ±1/4	Width of handle F ±1/16	Inside span with jaws in closed position G ±1/4	Thickness of handle H ±1/16	Rivet body diameter J ±1/16	Depth of jaw K ±1/4	Jaw opening, (min.)	Weight, (max.)
Inches	*Inches*	*Inches*	*Inch*	*Inches*	*Inches*	*Inch*	*Inches*	*Inch*	*Inch*	*Inch*	*Inch*	*Ounces*
6	6	1 1/8	5/8	1 3/4	1 1/2	1/4	1	11/32	1/4	11/16	1/2	10
8	8	1 3/8	11/16	2 1/4	1 3/4	5/16	1 3/8	3/8	5/16	13/16	3/4	16
10	10	1 5/8	3/4	2 7/16	2	3/8	1 5/16	3/8	5/16	15/16	13/16	22

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3.11 Workmanship. — Workmanship shall be first class in every respect.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Sampling for acceptance inspection.

4.2.1 *Inspection lot.*—All nippers or pincers of the same type, class, style, size, and having the same coating and offered for delivery at the same time shall be considered a lot for purposes of acceptance inspection.

4.2.2 *Sampling for visual and dimensional examination.*—A random sample of nippers or pincers shall be selected from each lot offered for Government inspection in accordance with Military Standard MIL–STD–105 at inspection level III for lots of 40 and under, inspection level II for lots of 41 to 300, and inspection level I for lots of 301 and over. The acceptable quality level (AQL) shall be 1.5 percent defective.

4.2.3 *Sampling for acceptance inspection.* A random sample of nippers or pincers shall be selected from each lot offered for Government inspection in accordance with the appendix to Military Standard MIL–STD–105 at inspection level L–7. The AQL shall be equal to 2.5 percent defective. However, all the sample sizes shall be used and the acceptance number shall be zero until the AQL permits one or greater.

4.3 Inspection.

4.3.1 *Visual and dimensional examination.*—Each of the sample nippers or pincers selected in accordance with 4.2.2 shall be visually and dimensionally examined to verify compliance with this specification. Any nipper or pincer in the sample containing one or more visual or dimensional defects, shall be rejected, and if the number of defective nippers or pincers in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3.2 *Acceptance inspection.*—Each of the sample nippers or pincers selected in accordance with 4.2.3 shall be tested in accordance with 4.4 to verify compliance with this specification. Any sample nipper or pincer which does not meet the requirements for any of these characteristics shall be rejected, and if the number of nonconforming nippers or pincers in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4 Test procedures.

4.4.1 *Hardness.*—The cutting edge hardness of all nippers and pincers shall be the mean value of three or more determinations taken in an area adjacent to the cutting edges. Samples ordinarily shall not be destroyed by fracturing in order to obtain suitable jaw surfaces for test. If reliable readings cannot be obtained on the finished nippers or pincers because of difficulty in supporting the test sample and maintaining the test surface level at the point of indentation, a suitable fixture shall be employed for clamping or properly supporting the sample under test. The jaws and handles of pincers and nippers furnished with a coated finish shall be prepared for the hardness test by grinding only sufficiently to insure accurate readings.

4.4.2 *Handle-load tests.*—The sample under test shall be supported in a manner suitable for applying the load properly and determining the deflection and permanent set of handles. Figure 8 shows a platform scale equipped for making load tests. The cutting edges of the jaws shall be in normal horizontal position, with the head resting on a support or guide. The lower handle shall rest on

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a suitably fixed knife edge support. Loads shall be applied through a suitable knife edge to the upper handle at a point in line with the lower handle support. For test loads given in pounds, the load specified herein shall be applied directly; for test loads given in inch-pounds, the load applied to the handle shall be calculated by dividing the test load specified herein by the distance between the center of the rivet and the line of action of the load. The width of the flat steel inserts shall be at least equal to that of the jaws. The method of determining the amount of deflection under load and the permanent set shall be that of marking the extreme ends of the handles with light centerpunch marks and measuring the distance between the marks by means of a divider and scale. After an initial application of a load of approximately 5 pounds, the distance between the punch marks shall be measured. The difference between the initial measurement and the measurement after the test load specified herein has been reduced to approximately 5 pounds shall be the permanent set. If necessary, the extreme ends of the handles may be ground to obtain a surface suitable for making the centerpunch marks. Load and deflection tests may be made on a universal testing machine, a suitably equipped platform scale or a special calibrated testing device employing dead weights. If the deflection under load or the permanent set is in excess of the maximum value specified herein or if the handles or jaws break, crack, chip, or spread under load, or if the rivet shows signs of weakness, the sample does not comply with this specification.

4.4.3 *Type I, end cutting nippers.*

4.4.3.1 *Class 1, detachable cutters.*

4.4.3.1.1 *Handle load test.*—With the adjusting stop screw removed and a piece of steel 0.05 inch thick, hardened to show a Rockwell number of 35 to 40 on the "C" scale, inserted between the jaws so as to contact over the full width, the deflection under load and permanent set of the handles shall not exceed the values specified in table VIII.

4.4.3.1.2 *Steel-wire-cutting test.*—The nippers shall make a total of nine cuts of uncoated single strand plow steel wire of the

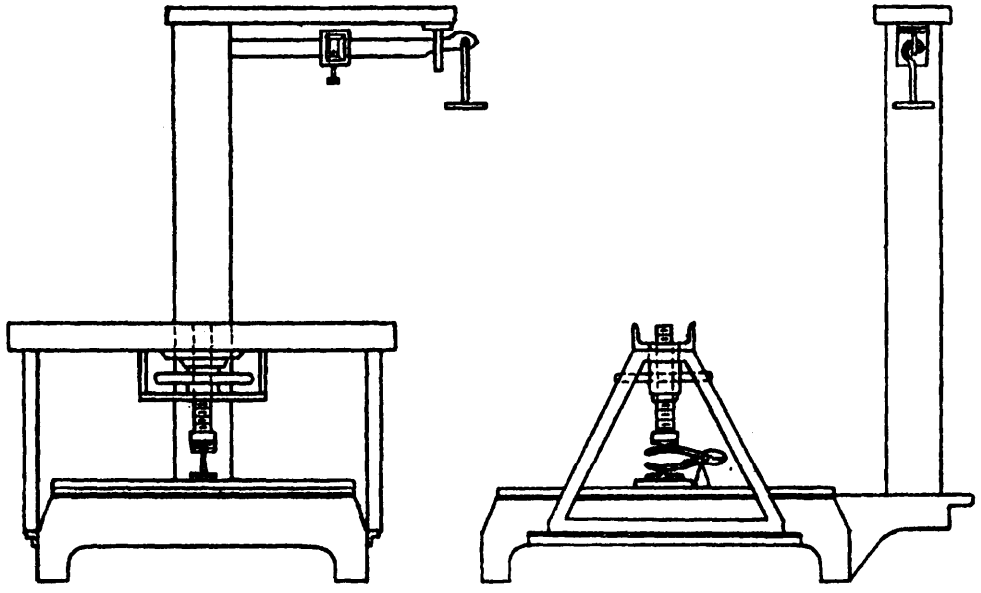


FIGURE 8.—Apparatus for applying test load to handles of pincers and nippers.

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TABLE VIII.—*Allowable deflection and set (type I, class 1)*

Nominal size	Test load	Point of load application from center of rivet	Deflection under load maximum	Permanent set maximum
Inches	*Pounds*	*Inches*	*Inch*	*Inch*
6	200	3½	0.50	0.08
7	75	5⅛	.400	.040
10	250	6½	.68	.06
12	275	8	.88	.09
14	350	9¼	1.00	.13

diameter specified in table IX. The steel stop screw shall be set so that the cutting edges just make contact. Three of the cuts shall be made at the approximate midpoint of the cutting edges and the test load required to completely sever the wire at this point shall not exceed the value specified in table IX. Three additional cuts shall be made at each end of the cutting edges with the wire located not more than ⅛ inch from the respective ends. Load determinations shall not be required for cuts made at the ends of the cutting edges.

TABLE IX.—*Wire diameter and test load for test cuts*

Nominal size	Wire diameter	Test load maximum
Inches	*Inch*	*Pounds*
6	0.080	100
7	.081	100
10	.091	150
12	.091	125
14	.091	125

4.4.3.2 *Class 2, integral cutter.*

4.4.3.2.1 *Style A, normal duty.*

4.4.3.2.1.1 *Handle-load test.*—With a piece of steel 0.05 inch thick, hardened to show a Rockwell number of 35 to 40 on the "C" scale, inserted between the jaws so as to contact over the full width, the deflection under load and permanent set of the handles shall not exceed the values specified in table X. The load shall be applied to the upper handle at the point of maximum handle curvature (the center of handle bow) and in line with the lower handle support.

TABLE X.—*Allowable deflection and set (type I, class 2, style A)*

Nominal size	Test load	Deflection under load, maximum	Permanent set, maximum
Inches	*Inch-pounds*	*Inch*	*Inch*
4½	500	0.38	0.03
6	1050	.50	.03
7	1500	.63	.03

4.4.3.2.1.2 *Steel-wire-cutting test.* — The nippers shall make a total of nine cuts of uncoated single strand plow steel wire of the diameter shown in table XI. Three cuts shall be made at the approximate mid-point of the cutting edges and the test load required to completely sever the wire at this point shall not exceed the value specified in table XI. Three cuts shall be made at each end of the cutting edges with the wire located not more than ⅛ inch from the respective ends. Load determinations shall not be required for cuts made at the ends of the cutting edges. Following completion of the nine cuts of wire, the cutter shall cut bond paper with the maximum load on the handles not exceeding that specified in table XI.

TABLE XI.—*Test load*

Nominal size	Wire diameter	Handle load, maximum
Inches	*Inch*	*Inch-pounds*
4½	0.062	500
6	.080	700
7	.091	850

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4.4.3.2.1.3 *Paper-cutting test.*—With a test load not exceeding 75 pounds on the handles, the nippers shall cut bond paper. Paper for cutting tests shall be 50 percent rag bond paper conforming to type B, substance 32 of Specification UU–P–121. The bond paper shall be in strip form, the width of the strip measuring 1/10 inch less than the length of the cutter edges. The strip of paper shall be centered between the ends of the cutting edges when the cut is made. Equipment and manner of applying load similar to that specified in 4.4.2 may be used. The paper cutting tests are intended to determine proper alignment of the cutting edges of the wire cutter. Paper cutting tests which follow wire-cutting tests are intended to determine whether deformation or failure has occurred due to crushing of the cutting edges of the wire cutter incident to cutting wire. The cutters of the nippers or pincers under test shall completely and cleanly cut the bond paper.

4.4.3.2.2 *Style B, light duty (jewelers') handle-load test.*—With a flat piece of steel 0.031 inch thick, hardened to show a Rockwell number of 35 to 40 on the "C" scale, inserted between the jaws so as to contact over the full width, and a test load of 80 inch-pounds applied 3 3/8 inches from center of screw, the handle deflection under load shall not exceed 0.40 inch and the permanent set of the handles shall not exceed 0.04 inch.

4.4.4 *Type II, pincers.*

4.4.4.1 *Class 1, farriers'.*

4.4.4.1.1 *Handle load test.*—With a piece of steel 0.05 inch thick, hardened to show a Rockwell number of 35 to 40 on the "C" scale, inserted between the jaws so as to contact over the full width, the deflection under load and permanent set of the handles shall not exceed the values specified in table XII.

4.4.4.1.2 *Steel-wire-cutting test.*—Class 1 pincers shall make a total of three cuts of uncoated single-strand plow-steel wire, 0.034 inch in diameter. One cut shall be made at the center and one cut at each end of the cutting edges within 1/8 inch from the end. The handle loads required to cut the wire shall not exceed the test loads specified in table XII.

TABLE XII.—*Allowable deflection and set (type II, class 1)*

Nominal size	Test load	Point of load application from center of rivet	Deflection under load, maximum	Permanent set, maximum
Inches	*Pounds*	*Inches*	*Inch*	*Inch*
8	144	5 1/4	0.63	0.06
10	160	6 1/2	.69	.06
12	172	8	.88	.09
14	200	9 1/4	1.00	.13

4.4.4.2 *Class 2, carpenters'.*

4.4.4.2.1 *Handle-load test.*—With a piece of steel 0.05 inch thick hardened to show a Rockwell number of 35 to 40 on the "C" scale inserted between the jaws so as to contact over the full width, the deflection under load and permanent set of the handles shall not exceed the values specified in table XIII.

TABLE XIII.—*Allowable deflection and set (type II, class 2)*

Nominal size	Test load	Point of load application from center of rivet	Deflection under load	Permanent set
Inches	*Pounds*	*Inches*	*Inch*	*Inch*
6	250	3 1/2	0.60	0.06
8	250	5 1/4	.63	.06
10	250	6 1/2	.69	.09

4.4.4.2.2 *Steel-wire-cutting test.*—Class 2 pincers shall make a total of three cuts of uncoated single-strand plow steel wire, 0.060

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inch in diameter for size 6 inches, 0.072 inch in diameter for size 8 inches, and 0.080 inch in diameter for size 10 inch pincers. One cut shall be made at the center and one at each end of the cutting edges within ⅛ inch from the end. The handle loads required to cut the wire shall not exceed the test loads specified in table XIII.

4.5 **Rejected lots.**—Rejected lots may be offered again for Government inspection provided the contractor has repaired or removed all nonconforming nippers or pincers. Samples shall again be selected from such resubmitted lots and inspected to verify compliance with this specification.

5. PREPARATION FOR DELIVERY

(Civil agencies.—Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.)

5.1 Cleaning, preservation, and packaging.

5.1.1 *Level A.* — Preservation and packaging shall be in accordance with level A of Military Specification MIL–P–15424.

5.1.2 *Level C.*—Cleaning, preservation, and packaging shall be in accordance with the contractor's commercial practice.

5.2 Packing.

5.2.1 *Level A.*—Nippers and pincers, packaged as specified (see 6.1), shall be packed in accordance with level A of Military Specification MIL–P–15424.

5.2.2 *Level B.*—Nippers and pincers, packaged as specified (see 6.1), shall be packed in accordance with level B of Military Specification MIL–P–15424.

5.2.3 *Level C.*—Nippers and pincers, packaged as specified (see 6.1), shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Shipping containers shall be in accordance with the rules or regulations of carriers as applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Military marking.* — In addition to any special marking required by the contract or order, or herein, interior and exterior shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

5.3.2 *Civilian marking.*—In addition to any special marking required by the contract or order, marking for shipment shall be in accordance with the standard marking of Federal Standard No. 123.

6. NOTES

6.1 **Ordering data.** — Procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type, class, and style of tool required (see 1.2).
(*c*) Size of tool required (see applicable table).
(*d*) Coating or combination required (see 3.7.2).
(*e*) Level of preservation and packaging, and packing required (see 5.1 and 5.2).
(*f*) Marking required (see 5.3).

6.2 When repair parts for type I, class 1, styles B and C nippers are required, the contract or order should give the following information: (a) manufacturer's name or trade-mark, (b) size of nippers, (c) name of parts and manufacturer's number or symbol, if available, and (d) any other data or description applicable to the nippers which will assist in identifying the proper repair parts.

6.3 **Superseding data.**—This specification supersedes Federal Specification GGG–P–471a in part covering nippers and pincers.

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The following types, classes, and styles of Federal Specification GGG–P–471a and this specification are equivalent:

Specification GGG–P–471a	*This specification*
Type A	Type I
Type A, style 1	Type I, class 1, style A
Type A, style 2	Type I, class 1, style B
Type B	Type I, class 2, style A
Type C	Type II, class 1

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY CUSTODIANS:
Army—Ordnance Corps
Navy—Bureau of Ships
Air Force.

Copies of this specification may be purchased for 10 cents each.

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